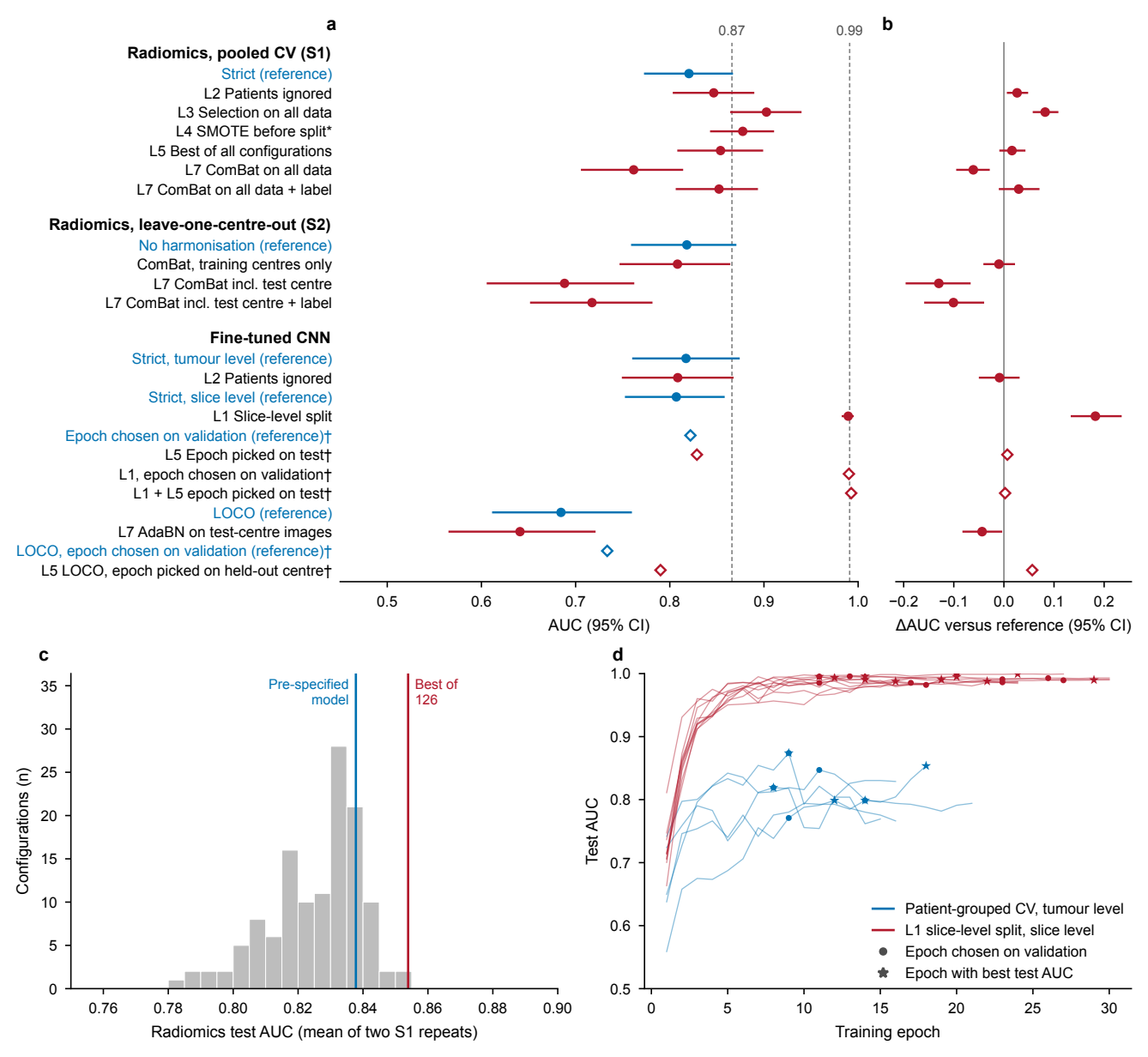

a
b
c
d
Radiomics, pooled CV (S1)
Strict (reference)
L2 Patients ignored
L3 Selection on all data
L4 SMOTE before split*
L5 Best of all configurations
L7 ComBat on all data
L7 ComBat on all data + label
Radiomics, leave-one-centre-out (S2)
No harmonisation (reference)
ComBat, training centres only
L7 ComBat incl. test centre
L7 ComBat incl. test centre + label
Fine-tuned CNN
Strict, tumour level (reference)
L2 Patients ignored
Strict, slice level (reference)
L1 Slice-level split
Epoch chosen on validation (reference)†
L5 Epoch picked on test†
L1, epoch chosen on validation†
L1 + L5 epoch picked on test†
LOCO (reference)
L7 AdaBN on test-centre images
LOCO, epoch chosen on validation (reference)†
L5 LOCO, epoch picked on held-out centre†
0.87
0.99
0.5
0.6
0.7
0.8
0.9
1.0
AUC (95% CI)
−0.2
−0.1
0.0
0.1
0.2
ΔAUC versus reference (95% CI)
Pre-specified model
Best of 126
Configurations (n)
0
5
10
15
20
25
30
35
0.76
0.78
0.80
0.82
0.84
0.86
0.88
0.90
Radiomics test AUC (mean of two S1 repeats)
Test AUC
0.5
0.6
0.7
0.8
0.9
1.0
0
5
10
15
20
25
30
Training epoch
Patient-grouped CV, tumour level
L1 slice-level split, slice level
Epoch chosen on validation
Epoch with best test AUC